Muhammad Nabeel Malik

Full-Stack AI Engineer | LLM Agents & Full Stack Development

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LinkedIn | GitHub | Portfolio

Professional Summary

Full-stack software engineer with 5+ years of experience building SaaS, fintech and e-commerce products, with a focus on real-time, data-heavy systems. Now working on AI engineering: LLM agents, RAG and tool calling in production. Main stack: Python, TypeScript, Nest.js, React.js, Next.js, PostgreSQL, Kafka, AWS and Docker.

Work Experience

Full-Stack AI Engineer (full-time), *OSPHIM* 07/2026 – Present
Aachen, Germany

- Moved the AI agent off a self-hosted Qwen setup onto the Anthropic Claude API and removed the old engine.
- Switched the main model from Claude Sonnet 4.6 to 5.5, which made each request about 13% cheaper. Small tasks stay on Haiku 4.5.
- Wrote the FastAPI backend and 12 tools for the agent. Operators can log defects, write handovers and edit assets from chat, and nothing is saved until they confirm it.
- Made the agent say when it cannot check something, such as "detection is not configured", instead of reporting "no anomaly". User PII never reaches the model.
- Set up evals with real questions from production to catch regressions before release.
- Owned the shift-handover module: shift plans, rostering, and handovers by chat or voice. It runs in 3 plants with 40+ shifts a week.
- Merged 56 PRs in NestJS, SvelteKit 5, PostgreSQL and Redis, with 250+ unit, integration and E2E tests.
- Cut dashboard load time on machine data from 1,150 ms to 16 ms by reading from roll-up tables instead of a 1.4-billion-row table.

Full-Stack Engineer (part-time), *OSPHIM* 01/2026 – 07/2026
Aachen, Germany

- Developed the first AI assistant on self-hosted Qwen models, a 35B model for chat and a 1.5B one for routing, with LlamaIndex agents and a Neo4j knowledge graph.
- Added voice handovers with self-hosted Whisper, including matching spoken machine names to the right asset.
- Rebuilt the production logbook so entries update live over WebSockets and can link 5+ entity types.
- Moved all logbook entries to JSONB with Drizzle ORM without any downtime.
- Added multi-cavity support to experiments. A 16-cavity mold now needs one setup, not 16.

Software Engineer, *MobileLive* 05/2024 – 10/2025
MonieCorp (Fintech Product) Remote

- Developed responsive and modular UI components in React + TypeScript + ShadCN, improving page load speed by 25% and reducing UI bugs by 40% after refactoring.
- Built 15+ SOLID-compliant React components for Admin panel using Cursor IDE for rapid development, cutting client feedback cycles from 3 days to 24 hours.
- Built multi-step React form for admins to configure tiered currency margins with validation; streamlined setup from 20+ minutes to <4 minutes per currency pair.

Staples Canada (MobileLive Client)

- Implemented Mixpanel tracking for 100,000+ users, capturing product views, add-to-cart, and checkout events, enabling business team to optimize flows and boost conversions by 12%.
- Built custom React features using Canva API for Staples Online Print Shop, enabling dynamic product personalization and increasing user engagement by 20%.
- Fixed 30+ production bugs within 4-hour SLAs, maintaining 99.9% uptime during peak sales.

Associate Software Engineer, *Code-Huddle* 08/2022 – 05/2024
GYMYG: Virtual Gym Partner (US-Based Startup) Islamabad, Pakistan

- Built Nest.js APIs with JWT RBAC, handling 10K DAU at <200ms P95 latency.
- Integrated Stripe monthly subscriptions, processing $15k+ in secure payments.
- Reduced video load times by approximately 30% using FFMPEG compression and optimized AWS CloudFront edge delivery.
- Automated end-to-end CI/CD pipeline with Github Actions, AWS CodePipeline, cutting deployment time by 40%.

Ordinami: QR Ordering Web-App (Italian Startup MVP)

- Implemented Firebase Auth + Firestore backend, supporting 100+ daily orders.
- Built an admin-panel shadcn/ui + Tailwind for menu creation, reducing manual setup time for owners by 60%.

Product Development Intern, *Code-Huddle* 04/2022 – 06/2022
Islamabad, Pakistan

- Assisted in frontend (React.js) and backend (NestJS) development, contributing to the design and implementation of scalable solutions.
- Supported AWS operations and collaborated on ongoing projects, gaining exposure to the full product development lifecycle.

Education

Bachelor of Science in Software Engineering, *COMSATS University* 2018 – 2022
Islamabad, Pakistan

Skills and Core Competencies

AI & LLM Engineering
Python | FastAPI | SQLAlchemy | Claude API | Qwen / Ollama | LlamaIndex | LLM Agents | Tool Calling | RAG | Evals | Whisper | Neo4j

Frontend Development
React.js | Next.js | SvelteKit | Context API | Tailwind CSS | HTML5 | CSS3 | Shopify (Liquid Templating) | ShadCN UI | Redux and Redux Toolkit | RxJs

Software Development & Engineering
Full-Stack Web Applications | RESTful APIs | Microservices Architecture | Secure Authentication Systems | Payment Integrations (Stripe) | CI/CD Automation | SaaS Product Engineering

Cloud, DevOps & Containerization
AWS (ECS, ECR, EC2, RDS, S3, CloudFront, CodePipeline, CloudWatch, SNS) | Azure | Docker

Backend Development
JavaScript | TypeScript | Node.js | Express.js | Nest.js | TypeORM | Prisma | Drizzle ORM | Kafka | Redis

Databases & Data Management
MySQL | PostgreSQL | MongoDB | Query Optimization | Data Modeling

Notable Freelance Projects

Pendfunds.com (M.V.P.), *Contract Management & Mediation Software* 01/2024 – 09/2024

- Developed full escrow payment flow with Stripe Connect and Optimus Bank merchant payouts in Nest.js, onboarding 100+ users and handling hundreds of transactions.
- Automated deployments across development, staging, and production environments using AWS Elastic Beanstalk and AWS CI/CD pipelines.
- Implemented secure payload encryption/decryption using PGP with RSA-256, ensuring sensitive transaction data remained fully protected.